

1. Aim

The aim of this project is to develop an interactive Power BI sales dashboard that presents sales data in a clear and simple way. The dashboard helps users understand sales, profit, customers, products and regional performance through interactive visualisations and filters.

2. Objectives

The main objectives of the project are:

- To analyse the sales dataset and understand the information available.
- To calculate important business measures such as total sales, total profit and total orders.
- To identify the top-performing products and customers.
- To analyse sales and profit across different regions and categories.
- To identify sales and profit trends over time.
- To use suitable visualisations to present the data clearly.
- To create interactive slicers and page navigators for easier dashboard navigation.
- To provide useful insights and recommendations based on the analysis.
- To create a user-friendly dashboard that helps users explore the sales data and support business analysis.

3. Data Description

The Power BI dashboard is created using a sales dataset. The data contains information about customer orders, products, sales, profit and regions.

Main Data Fields

Field	Description
OrderID	Unique identification number for each order
OrderDate	Date on which the order was placed
Product	Name of the product purchased
CustomerName	Name of the customer
Category	Category of the product
Region	Region where the order was made

Quantity	Number of units ordered
Sales	Total sales/revenue generated
Profit	Profit generated from the order

Use of the Data

The data is used to analyse:

- Total number of orders
- Total sales
- Total profit
- Profit margin
- Product performance
- Customer performance
- Regional performance
- Quantity sold
- Sales and profit trends over time

4. Data cleaning

- Missing value handling
 - Customer name, region, product, category-Replace with unknown
 - Order date missing columns (2 rows deleted)-Deletion
 - Unit price, sales, cost, profit-Imputation with median
- Duplicate handling
 - 18 duplicate records found hence deleted 18 rows., remaining 1000 rows.
- Verified column data types (Text, Whole Number, Decimal Number, Date) and corrected where necessary.

5. Power BI Dashboard Visualisation Explanation

Page 1 – Executive Summary

The first page provides an overall summary of the sales data. It helps users quickly understand the company's sales, profit, orders and customer performance.

Visualisations Used

1. KPI Cards

KPI cards are used to display important business figures such as:

- Total Orders
- Total Sales
- Total Profit
- Profit Margin
- Average Order Value

They provide a quick summary of the overall business performance.

2. Column Chart – Top 5 Products

This visualisation shows the top five products based on the number of orders. It helps identify which products are ordered most frequently.

3. Line Chart – Profit Trend

The line chart shows how profit changes over time. It is useful for identifying increases, decreases and trends in profitability.

4. Bar Chart – Top 5 Customers

This chart shows the customers with the highest number of orders. It helps identify the customers who contribute most to order volume.

5. Line Chart – Sales Trend

This visualisation shows sales performance over time and helps identify changes in sales during the reporting period.

6. Pie Chart – Orders by Region

The pie chart shows the proportion of orders received from different regions. It helps compare regional order distribution.

7. Slicers

Slicers are used for interactive filtering, such as Category, Product and Quantity. Users can select values and the dashboard updates automatically.

Page 2 – Sales and Distribution Analysis

The second page provides a more detailed analysis of products, sales, profit, regions and quantities.

Visualisations Used

1. Bar Chart – Product vs Sales

This chart compares the sales generated by different products. It helps identify products that generate higher sales.

2. Column Chart – Product vs Profit

This chart compares the profit generated by different products. It helps identify which products contribute more to profitability.

3. Donut Chart – Category/Region Distribution

The donut chart shows how orders are distributed across different categories and regions. It provides an easy way to compare proportions.

4. Map – Sales and Profit by Region

The map shows sales and profit geographically. It helps users understand which regions contribute to business performance.

5. Bar Chart – Quantity by Product

This visualisation compares the quantity sold for different products. It helps identify products with higher demand in terms of quantity.

6. Slicers

Slicers are provided for fields such as Customer, Region, Sales and Profit. They allow users to filter the dashboard and perform more detailed analysis.

Page 3 – Insights and Recommendations

The third page is used to present the main findings from the sales analysis.

Purpose of the Page

This page summarises the important observations obtained from the first two dashboard pages. It helps management understand the main patterns in sales, profit, customers, products and regions.

Information Included

- Key sales findings
- Profit-related findings
- Top-performing products
- Important customers
- Regional performance
- Trends observed in sales and profit
- Recommendations based on the dashboard results

This page converts the information from the visualisations into simple business insights and recommendations.

Page Navigators

Page Navigators are used to help users move easily between the different pages of the Power BI dashboard.

In this dashboard, the page navigator allows the user to move between the three main pages:

1. Executive Summary – Provides an overall view of sales, profit, orders and customers.

2. Sales & Distribution Analysis – Provides detailed analysis of products, sales, profit, quantity and regions.
3. Insights & Recommendations – Presents the main findings and recommendations from the analysis.

6. DAX Calculations Performed

- Created a Calculated Table named *EastRegionOrders* using a filter on the Region column.
- Added a Calculated Column named *ProfitCostDifference* to compute the difference between Profit and Cost.
- Developed a Measure named *TotalProfit* to calculate the sum of all profit values.
- Validated DAX outputs by cross-checking values with the dataset and visuals.